Statistical Machine Learning — Course Project

Text Classification: Sparse Features vs Static and Contextual Embeddings

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1 Introduction and Dataset

This project compares classical ML models across three text representation paradigms: *sparse features* (BoW, TF-IDF), *static dense embeddings* (GloVe), and *contextual embeddings* (DistilBERT). Only classical models are trained; DistilBERT and GloVe are used strictly as frozen feature extractors. The three-way axis isolates whether gains come from density alone or from contextual understanding.

Dataset. AG News via Hugging Face, 8,000 samples (seed = 42), 4 balanced classes (*World, Sports, Business, Sci/Tech*, ≈2,000 each). Split: 6,400 train / 1,600 test (80/20, random_state = 42).

2 Dataset Description and EDA

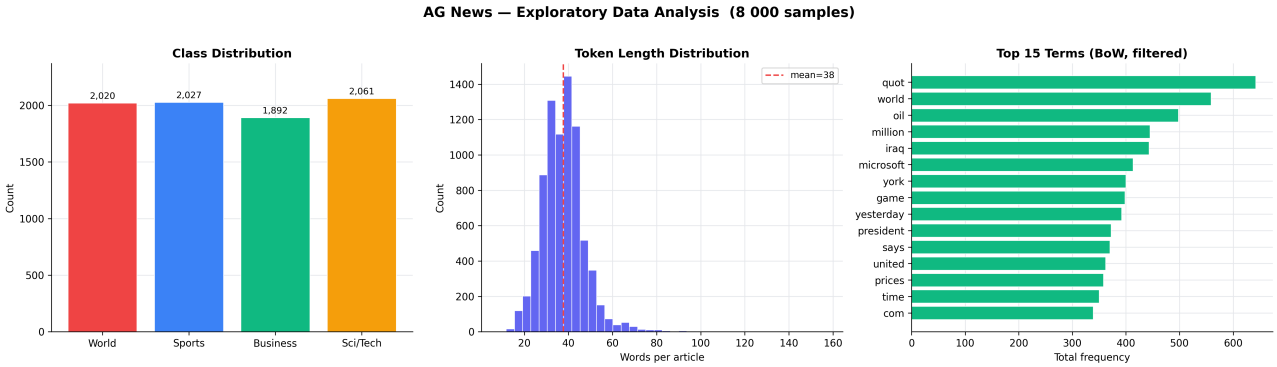


Figure 1: EDA: class distribution (balanced), token-length histogram (peak 40–45 words), and top-15 BoW terms after stop-word removal.

The dataset is near-perfectly balanced, eliminating class-imbalance as a confound. Article lengths follow a roughly Gaussian distribution; very few exceed 100 tokens. Discriminative vocabulary (extracted from LR coefficients) shows Sports vocabulary is highly domain-specific, while Business and Sci/Tech share technological terminology, foreshadowing their classifier confusion.

3 Methodology

Part A: Sparse Features.

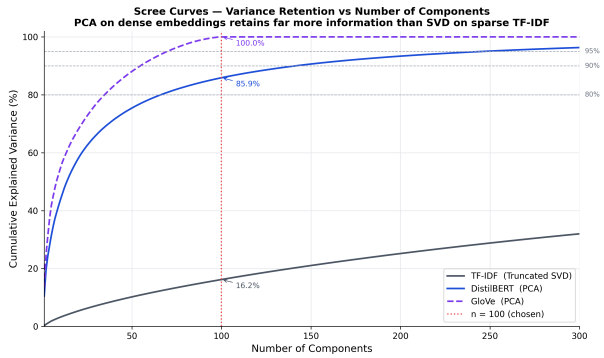
- **BoW**: unigram raw-count, top-5,000 vocab, min_df = 2, alphabetic tokens ≥3 chars.
- **TF-IDF**: unigram+bigram, sublinear_tf, top-5,000 vocab, min_df = 2.

Part B: GloVe (Bonus). 100-d glove-wiki-gigaword-100 (6B tokens); documents represented as arithmetic mean of known word vectors, cached as `.npy`.

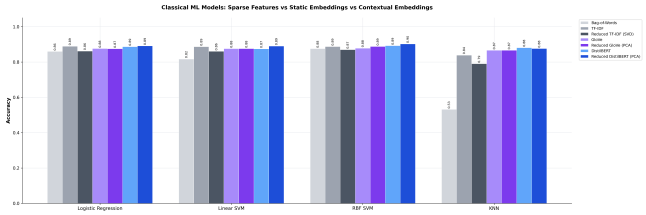
Part C: DistilBERT. Frozen `distilbert-base-uncased`; [CLS] token from final hidden state yields a 768-d vector per article, computed once on GPU (Kaggle) and cached.

Part D: Models. Logistic Regression (`max_iter=1000`), Linear SVM, RBF SVM, KNN ($k = 5$), KMeans ($K = 4$, unsupervised baseline).

Part E: Dimensionality Reduction. Truncated SVD on TF-IDF and PCA on DistilBERT/GloVe, all compressed to 100 components.



(a) Scree curves: PCA retains 85.89% of DistilBERT variance at n=100; SVD retains only 16.19% for TF-IDF.



(b) Accuracy across all models and feature spaces. Sparse → GloVe → DistilBERT is consistent.

Figure 2: Dimensionality reduction variance retention (left) and overall accuracy comparison (right).

4 Experiments

Each supervised model was trained on an 80/20 stratified split (`random_state=42`) and evaluated on held-out accuracy and macro F1. KMeans was fit on the full dataset and evaluated via NMI and ARI. Dimensionality reduction was applied independently per feature space (100 components).

5 Results

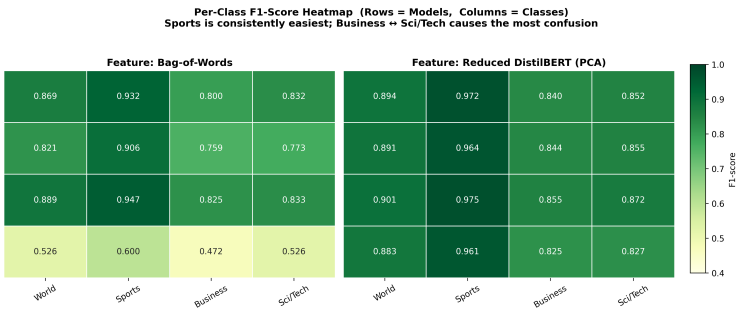
5.1 Full Numerical Results

Table 1: Accuracy and Macro F1-score for all model–feature combinations. Best per model in **bold**.

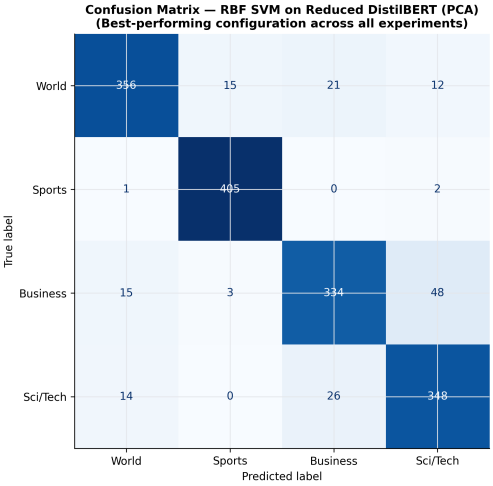
Model	Feature Set	Accuracy	Macro F1
Logistic Regression	Bag-of-Words (unigram)	0.8594	0.8582
Logistic Regression	TF-IDF (unigram+bigram)	0.8888	0.8874
Logistic Regression	Reduced TF-IDF (SVD, 100d)	0.8612	0.8596
Logistic Regression	GloVe mean-pool (100d)	0.8756	0.8745
Logistic Regression	Reduced GloVe (PCA, 100d)	0.8744	0.8733
Logistic Regression	DistilBERT (768d)	0.8862	0.8854
Logistic Regression	Reduced DistilBERT (PCA, 100d)	**0.8906**	0.8895
Linear SVM	Bag-of-Words (unigram)	0.8163	0.8145
Linear SVM	TF-IDF (unigram+bigram)	0.8862	0.8851
Linear SVM	Reduced TF-IDF (SVD, 100d)	0.8600	0.8585
Linear SVM	GloVe mean-pool (100d)	0.8750	0.8739
Linear SVM	Reduced GloVe (PCA, 100d)	0.8750	0.8739
Linear SVM	DistilBERT (768d)	0.8738	0.8723
Linear SVM	Reduced DistilBERT (PCA, 100d)	**0.8894**	0.8884
RBF SVM	Bag-of-Words (unigram)	0.8750	0.8737
RBF SVM	TF-IDF (unigram+bigram)	0.8869	0.8856
RBF SVM	Reduced TF-IDF (SVD, 100d)	0.8694	0.8682
RBF SVM	GloVe mean-pool (100d)	0.8775	0.8766
RBF SVM	Reduced GloVe (PCA, 100d)	0.8875	0.8866
RBF SVM	DistilBERT (768d)	0.8919	0.8908
RBF SVM	Reduced DistilBERT (PCA, 100d)	**0.9019**	0.9009
KNN	Bag-of-Words (unigram)	0.5312	0.5309
KNN	TF-IDF (unigram+bigram)	0.8381	0.8361
KNN	Reduced TF-IDF (SVD, 100d)	0.7900	0.7876
KNN	GloVe mean-pool (100d)	0.8656	0.8640
KNN	Reduced GloVe (PCA, 100d)	0.8656	0.8640
KNN	DistilBERT (768d)	**0.8806**	0.8792
KNN	Reduced DistilBERT (PCA, 100d)	0.8756	0.8741

KMeans (unsupervised): BoW NMI = 0.0452, ARI = 0.0142 TF-IDF NMI = 0.2411 DistilBERT NMI = 0.5589, ARI = 0.5727 GloVe NMI = 0.4504, ARI = 0.3840.

5.2 Per-Class F1 and Error Analysis



(a) Per-class F1: BoW (left) vs Reduced DistilBERT PCA (right). Sports easiest; Business/Sci/Tech confusion persists.



(b) RBF SVM on Reduced DistilBERT (PCA) (90.19%). Sports 405/408; primary error: 48 Business→Sci/Tech.

Figure 3: Per-class F1 heatmap (left) and best-model confusion matrix (right).

6 Discussion

Curse of dimensionality. KNN on 5,000-d sparse BoW scores only 53.12%; on dense 768-d DistilBERT it reaches 88.06% which is a 35-point gain from feature change alone.

Compression behaviour. PCA retains 85.89% of DistilBERT variance in 100 components, allowing LR to *improve* (88.62% → 89.06%). SVD on TF-IDF retains only 16.19%, causing a moderate drop (88.88% → 86.12%). Compression is safe for structured dense embeddings; destructive for isotropic sparse representations.

GloVe as semantic bridge. RBF SVM reaches 87.75% on GloVe vs 90.19% on Reduced DistilBERT. The gap isolates the contribution of *contextual* disambiguation, not density. The full hierarchy is: BoW < TF-IDF < GloVe < DistilBERT, mapping directly to no-semantics → co-occurrence → static word semantics → contextual sentence semantics.

Class-level confusion. Sports is the easiest class due to highly domain-specific vocabulary. 48 Business articles are misclassified as Sci/Tech as modern tech-company news genuinely straddles both domains. This confusion hierarchy is preserved across all feature spaces.

KMeans as unsupervised probe. DistilBERT achieves NMI = 0.5589 and ARI = 0.5727 *without any classifier*, vs NMI = 0.0452 for BoW likewise confirming the class separation is a geometric property of the embedding space, not a classifier artefact.

Limitations. Single 80/20 split (no cross-validation for main results); default sklearn hyperparameters throughout; GloVe mean-pooling ignores word order; DistilBERT CLS token only (mean-pooling often more robust).

7 Conclusion and Learnings

The best configuration is **RBF SVM on Reduced DistilBERT (PCA)** at **90.19%** (Macro F1 = 90.09%). Classical ML performance is bounded by input feature quality: provide a good feature space and even simple models excel. PCA on DistilBERT is information-preserving (85.89% retention) and actually improves downstream accuracy by removing noise; SVD on TF-IDF is destructive (16.19% retention) because sparse representations lack low-rank structure. DistilBERT's advantage over GloVe is attributable to contextual disambiguation, not mere density, as the GloVe baseline eliminates density as a confound.